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SOUND SOURCE LOCALIZATION

SYSTEM DESIGN DOCUMENT

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VERSION HISTORY

Version	Date	Author	Changes
0.1	11-04-2024	Casey Stijlaart	Layout, Introduction, Description
0.2	12-04-2024	Casey Stijlaart	System design (use case and requirements)
0.3	15-04-2024	Casey Stijlaart	System Design (use cases, requirements), Old Class Diagram Teensy
0.4	10-06-2024	Casey Stijlaart	Redefine the use cases and components, due to adjustments in scope.
0.5	14-06-2024	Casey Stijlaart	Completed class diagram for Teensy and common. Start sequence diagrams.

INTRODUCTION

This document functions as a detailed guide for the design and architecture of the embedded system tasked with localizing sound sources. Within its pages, readers will find an exhaustive system description, accompanied by a various diagrams, which define the system architecture and Behaviour.

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1 SYSTEM DESCRIPTION

The system is engineered to accurately localize multiple sound sources and visually represent them on a 2D map through an intuitive user interface.

1.1 EXTENSIONS

Must

- The system must possess the capability to detect and a sound source.
- A basic and straightforward test setup must be implemented for the prototype.
- The system must have a user friendly user interface.

Should

- Optimal microphone positioning techniques should be utilized to enhance measurement accuracy.
- Localized sound sources should be accurately represented on the map, with a maximum positioning error of 10%.

Could

- The prototype could be designed with flexibility for potential extensions or upgrades.
 - ▶ Possible extensions include integrating additional detection or processing devices for enhanced functionality.

2 SYSTEM DESIGN

Behaviour[System Design]

The system design section presents an overview of the architectural approach and components that constitute the system. It outlines the structural framework and technical aspects governing the system's functionality and scalability.

2.1 COMPONENTS

To ensure the system is working as intended, there are various components which work together. Each component has their own set of tasks, which they must perform. These components are described in

Component	abbreviation	Description
Audio Processor	AP	The audio processor unit takes care of processing all the collected raw microphone data
Communication Processor	CP	The communication processor serves as the communication bridge between the server and the audio processor units.
Server (UI)	SV	The server's user interface is the tangeable system part for the user. The gathered processed data is stored and displayed here, and user input can be given via here as well.

2.2 USE CASES

Use cases provide a detailed understanding of how the system functions in real-world scenarios. Each use case represents a specific interaction or functionality within the system, illustrating its Behaviour and purpose.

ID: UC_01	Use Case Name: Localize Sound Sources
Discription	Whenever the system detects any sound, the signals get picked up by the various microphones and are used to locate where the sound came from.
Actor(s)	AP, SV

ID: UC_02	Use Case Name: Real-time Visualization
Discription	The system should provide real-time visualization of detected sound sources on the map, updating the positions as new data is processed.
Actor(s)	User, UI

2.3 REQUIREMENTS

This section outlines the various requirements that the system must meet to fulfill its intended functionality. These requirements are categorized into User Requirements, Functional Requirements, and Non-Functional Requirements, each serving a specific purpose in defining the system's capabilities and characteristics.

2.3.1 USER REQUIREMENTS

User requirements are statements in natural language plus diagrams that describe the features and functionalities of the system from the perspective of its users.

ID	Requirement	Use Case(s)
UR01	The system interface must be intuitive and user-friendly.	UC_02
UR02	Users should be able to visualize localized sound sources in real-time on the map interface.	UC_01

2.3.2 FUNCTIONAL REQUIREMENTS

Functional requirements specify the Behaviour of the system, including its inputs, outputs, and the actions it should take in response to various inputs. These requirements detail what the system must do to fulfill the user requirements.

ID	Requirement	UC / UR
FR01	The system must be capable of detecting and localizing multiple sound sources simultaneously.	UC_01, UR_02
FR02	A basic test setup must be implemented to verify the prototype's functionality.	UC_02, UR_01

2.3.3 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements specify the criteria that can be used to judge the operation of a system, rather than specific Behaviours. These requirements describe system attributes such as performance, usability, reliability, and security.

ID	Requirement
NFR01	The system should demonstrate high accuracy in sound source localization under various environmental conditions.
NFR02	The system's user interface should be responsive and provide real-time updates of localized sound sources.
NFR03	The system should be easily scalable to accommodate additional sound detection devices or processing units.
NFR04	The system's components and communication protocols should ensure minimal latency in sound localization and visualization.

3 SYSTEM BEHAVIOUR

The system is organized into various sections, each serving specific targets. These sections are divided into packages for class definitions and will be separated by folder structure in the implementation.

The sections can be outlined as following :

- **Embedded**
 - ▶ **Common:** This package contains common definitions for the various targets on the embedded side.
 - ▶ **Teensy:** This package contains target-specific definitions for the Teensy target.
 - ▶ **ESP32:** This package contains target-specific definitions for the ESP32 target.
- **Server:** This package contains the definition for the server side.

3.1 SEQUENCE DIAGRAMS

Scenario 1: Initiate Startup

Regarding to the first scenario, the various used variables have the following values.

```
uart_data_in = 'S' 'L' 0x01 0x02 0x0E
return_val = 0x00
buffer_size = 0x01
uart_data_out = 'S' 'L' 0x01 0x82 0x87
```

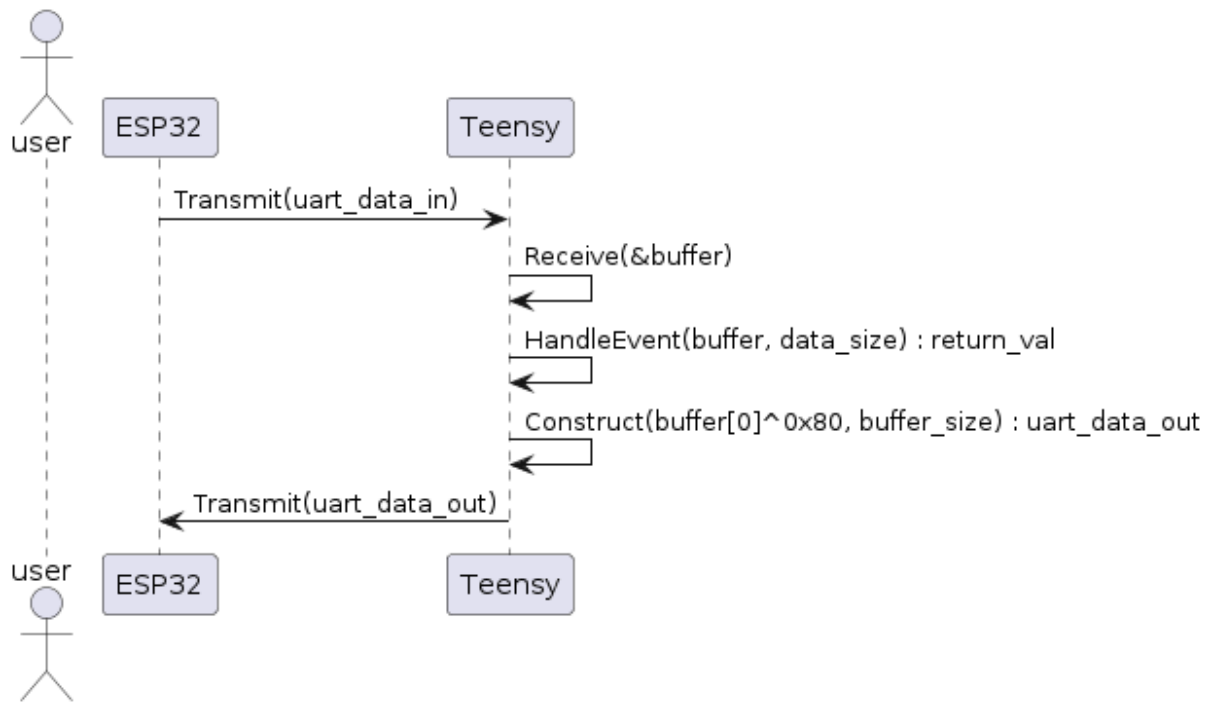


Figure 1: Scenario 1 : Initiate startup

Regarding to the second scenario, the various used variables have the following values.

[illegible]

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Scenario 3: Teensy Error

```
uart_data_in = 'S' 'L' 0x01 0x78 0x6F
return_val = 0x01
buffer_size = 0x02
uart_data_out = 'S' 'L' 0x02 0xEE 0x01 0x92
```

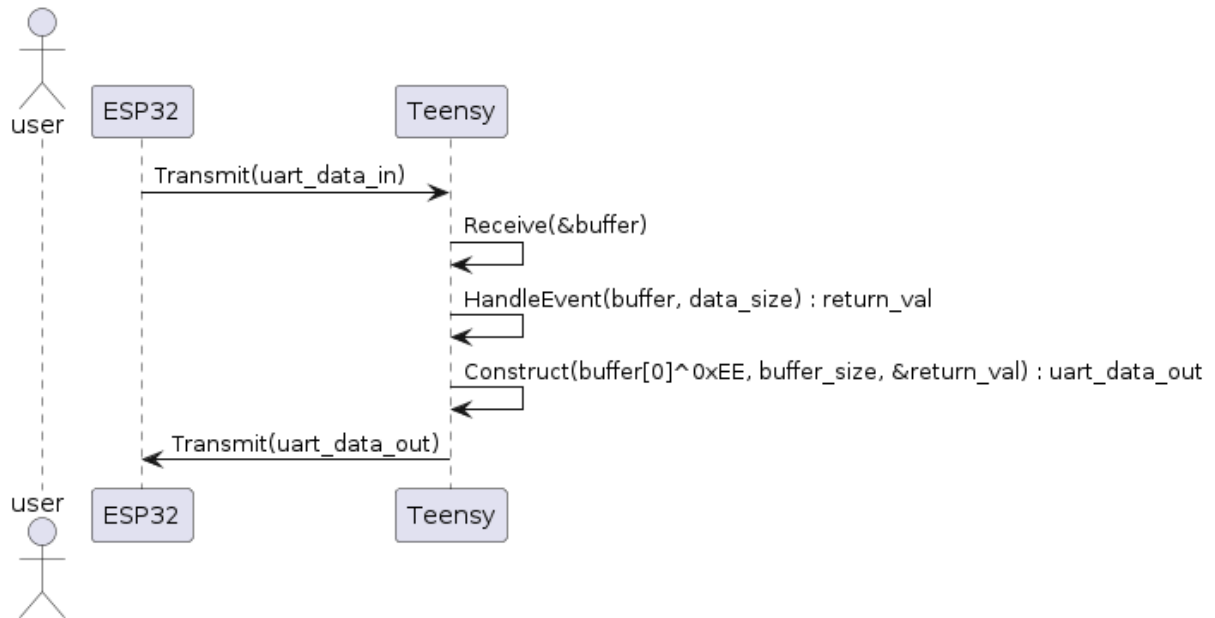


Figure 3: Scenario 3 : Error occurs on teensy upon request

3.2 CLASS DIAGRAMS

For clarity, each package is showcased separately. However, to illustrate the dependencies between some sub-packages, the following overview is provided. See Figure 4.

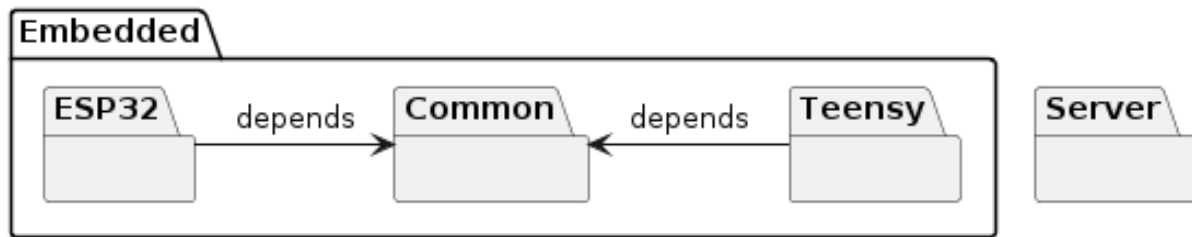


Figure 4: The relations between the various, so called, packages.

3.2.1 EMBEDDED : COMMON

The following figure displays abstractly how the classes within the Common package are connected.

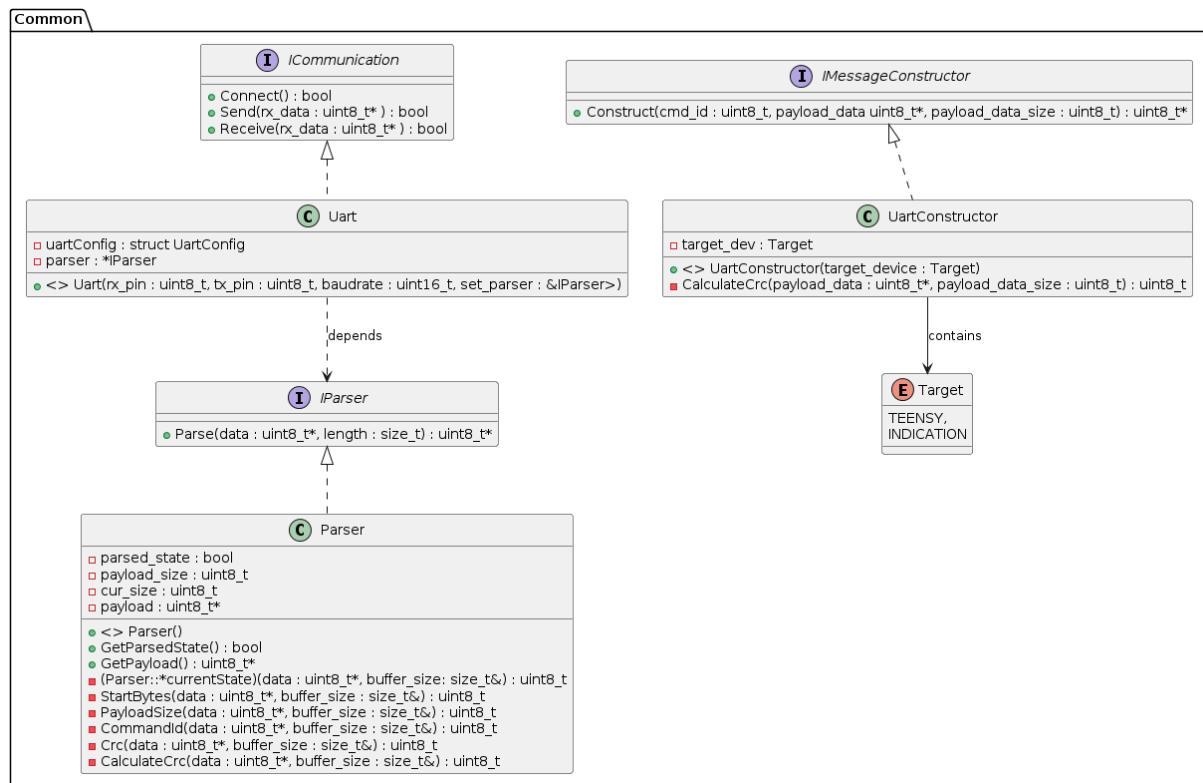


Figure 5: The class diagram for the package Embedded - Common.

3.2.2 EMBEDDED : TEENSY

The following figure displays abstractly how the classes within the Teensy package are connected.

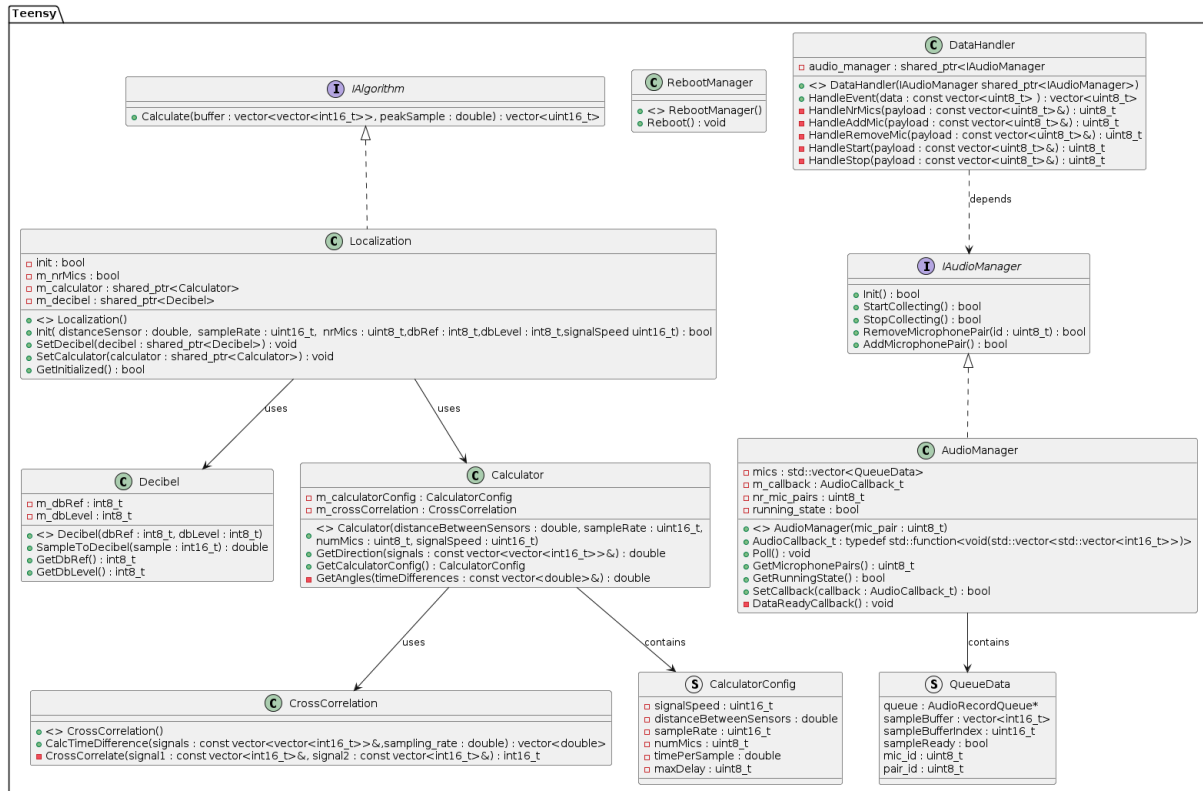


Figure 6: The class diagram for the package Embedded - Teensy.

3.2.3 EMBEDDED : ESP32

The following figure displays abstractly how the classes within the ESP32 package are connected.

3.2.4 SERVER

The following figure displays abstractly how the classes within the Server package are connected.

4 HARDWARE DIAGRAM

< Hardware Diagram >

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